IBTISSAM TOURE

Computer Science Student | AI Researcher | Technical Lead

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EDUCATION

Bachelor of Computer Science — Arab Open University, Riyadh*Expected 2028*
GPA: **3.68 / 4.00** • **Dean's List** (2026)

RESEARCH EXPERIENCE

Research Intern — Pedestrian Attribute Recognition — KFUPM & SDAIA Joint Research Center for Artificial Intelligence (JRCAI)*June 2026 – Present*
Project: Visibility-Guided Multimodal Knowledge Graph (VGMKG) for Pedestrian Attribute Recognition from Drone-Captured Imagery • *Supervisor: Dr. Hussein Samma*

- **Designed and implemented** a multimodal computer vision pipeline combining independent DINOv2 ViT-B/16 encoders (global and region-level crops), a graph neural network over a knowledge graph of region/part/attribute nodes, and Sentence-BERT text embeddings for human attribute recognition from drone-captured pedestrian imagery.
- **Built a skeleton-based visibility detection module** that consumes pre-computed COCO-style keypoint data to reason about occluded or out-of-frame body regions, replacing an earlier MediaPipe-based approach directed by the project supervisor.
- **Iterated on the model architecture** through several supervisor-directed revisions, including moving from GATv2Conv to a Graph Transformer attention layer and restructuring classification into per-group multi-layer branches.
- **Diagnosed and corrected a data leakage issue** caused by random splitting of augmented images from the same individuals, and adopted an identity-disjoint evaluation protocol; identified and fixed additional bugs across the loss computation, graph construction, and evaluation code.
- **Designed a structured ablation study** (baseline, text-only, visual-only, GNN-ablated, and full model) to isolate the contribution of each modality and architectural component. Full results are being finalized as part of the paper.
- **Conducted a literature review** spanning 80 papers across 8 categories in the pedestrian attribute recognition space, producing per-paper relevance assessments and a citation roadmap used to position the project's contribution.
- **Co-authoring an IEEE-format conference paper** on this work (currently in preparation); also a co-author on a second research paper within the same program that is likewise in progress.

TECHNICAL LEADERSHIP

Technical Lead — Volunteer — World Assembly of Muslim Youth (WAMY)*August 2026 – Present*

- **Lead the technical direction** of volunteer projects, translating community needs and project requirements into concrete technical solutions.
- **Coordinate technical development workflows** across project teams and contribute to platform and product design decisions for web/software solutions supporting social-impact and educational initiatives.
- **Guide projects from concept to functioning prototype**, including the Darsak Alayna nonprofit tutoring platform described under Projects.

MACHINE LEARNING EXPERIENCE

Machine Learning Intern — FlyRank*July 2026 – Present*

- **Working within an applied ML track** focused on real-world search and content analytics data, covering data wrangling, semantic embedding generation, and clustering to surface patterns in large, unstructured datasets.
- **Developing intent classification and opportunity-scoring approaches** that connect model output to actionable, business-relevant recommendations.
- **Using a reproducible, notebook-based ML workflow** in Python with pandas, scikit-learn, and sentence-transformers, applying clustering methods (e.g., HDBSCAN) for semantic grouping and analysis.

PROJECTS

Munsit — Real-Time Arabic Accessibility System*June 2026*
github.com/IbtissamToure/Munsit

- **Designed and built** a real-time, multimodal AI system that unifies speech recognition, sign-language detection, and environmental audio classification into a single pipeline to support deaf and hard-of-hearing users in Arabic-speaking environments.
- **Implemented Arabic speech-to-text** using OpenAI Whisper, and trained a fully connected sign-language classifier (63 → 128 → 64 → 28) on MediaPipe hand landmarks, reaching **84.2% validation accuracy** across 28 classes on a 4,507-sample Arabic Sign Language dataset.

- **Trained a CNN-based environmental sound classifier** on Mel-spectrogram audio features (UrbanSound8K, 5 classes), reaching **88.5% validation accuracy**, used to alert users to safety-relevant environmental sounds.
- **Deployed the system** to Streamlit Community Cloud with an integrated text-to-speech output layer for spoken sign predictions.

ShiftGuard — Shift Safety Risk-Scoring Tool*March 2026*
github.com/IbtissamToure/ShiftGuard

- **Built a decision-support tool** that analyzes employee shift schedules against the ILO's 11-hour minimum rest standard, computing a 0–100 risk score per employee and flagging dangerous scheduling patterns.
- **Designed the scoring and recommendation logic** in Python and pandas, translating each flagged pattern into a concrete, ranked replacement suggestion rather than a raw score alone.
- Presented at GOSHATHON 2025, motivated by real fatigue-related incidents among shift workers.

Darsak Alayna (درسك علينا) — Nonprofit Tutoring Platform Prototype*August 2026*
github.com/IbtissamToure/darsak-alayna

- **Developed a bilingual (Arabic/English) interactive prototype** for a nonprofit platform connecting students who cannot afford private tutoring, or who face medical circumstances, with volunteer teachers for free live weekly classes.
- **Designed end-to-end flows** for student registration, teacher applications and selection, class scheduling, attendance and guardian notifications, and teacher ratings — spanning student, teacher, parent, and admin roles.
- Built with React; supported by the World Assembly of Muslim Youth (WAMY). Currently a front-end prototype, not yet in production.

Drowsiness Detection — Driver Drowsiness Detection System*December 2025*
github.com/IbtissamToure/drowsiness-detection

- **Built a computer vision pipeline** to detect driver drowsiness from webcam video using transfer learning on MobileNetV2 for image-based classification.
- Implemented the full experimentation workflow — dataset preparation, exploratory analysis, model training, evaluation, and real-time webcam testing — and documented the impact of compute and data constraints on model performance.

TECHNICAL SKILLS

Programming: Python, JavaScript, HTML, CSS

AI / Machine Learning: Machine Learning, Deep Learning, Computer Vision, NLP, Transfer Learning

Computer Vision: Image Classification, Human Attribute Recognition, MediaPipe, OpenCV

Multimodal / Graph AI: Vision Transformers (ViT, DINOv2), Knowledge Graphs, Graph Neural Networks (Graph Transformer, GATv2Conv), Sentence-BERT, Multimodal Fusion

Data / ML Engineering: PyTorch, TensorFlow, pandas, scikit-learn, sentence-transformers, Jupyter, Data Preprocessing, Model Evaluation, HPC/SLURM

Development: Git, GitHub, React, Streamlit, Full-Stack Development, Web Development

Research: Data Annotation, Experimental Design, Ablation Studies, Literature Review, Academic Writing (LaTeX/IEEEtran)

ACHIEVEMENTS & ACTIVITIES

- **Dean's List**, Arab Open University (2026)
- **180 Seconds of Impact Competition** — Participant, AOU Annual Scientific Research Forum 2025–2026
- **Basira** (AI-powered AR smart glasses concept for hearing-impaired students) — presented, Creative Ideas Category, AOU Annual Research Forum 2026
- **Certifications:** McKinsey Forward Program; Microsoft Azure AI Fundamentals (AI-900); Huawei HCIA-AI V3.5; AWS Machine Learning Foundations; IBM Machine Learning with Python; University of Michigan — Python for Everybody (4-course specialization)
- Member, AI Club and Google Developers Club — technical and educational volunteering